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Zero Click Marketing and the Measurement of Distributed Influence

Evidence from Search, Social Platforms, Generative AI, and Google Marketing Live 2026

Emilio Alcolea

Founder, Tersefy

Tijuana, Baja California, Mexico

alcolea@tersefy.com

ORCID: [0009-0003-0666-2232](https://orcid.org/0009-0003-0666-2232)

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Abstract

This working paper defines Zero Click Marketing as a strategy and measurement framework for environments in which exposure, comprehension, recall, trust, and future demand are formed without a visit to brand-owned properties. It synthesizes evidence from zero-click search studies, social-platform ranking behavior, generative answer engines, cognitive psychology, brand-effectiveness research, and Google Marketing Live 2026.

The paper argues that clicks remain valid demand-capture signals but are insufficient as the sole proxy for marketing influence. A growing share of influence is formed inside platforms, AI-mediated answer surfaces, communities, and private channels before a measurable click occurs. Google Marketing Live 2026 strengthens the directional thesis by introducing platform-owned measurement products such as Attributed Branded Searches and Qualified Future Conversions, while also raising a new methodological concern: the successor metrics to click attribution are being developed primarily by the platforms that benefit from controlling them.

The contribution is a conservative synthesis rather than a causal proof. The paper proposes a Zero Click Ladder, separates attribution from measurement, classifies

claims by evidence tier, and outlines a research agenda for testing distributed influence across search, social, AI answer engines, and private-channel environments.

Keywords: zero-click search, distributed influence, native content, generative engine optimization, mental availability, attribution, media mix modeling, dark social, brand salience, agentic commerce, primary-source evidence.

JEL classification: M31 (Marketing), M37 (Advertising and Product Promotion), L86 (Information and Internet Services), D83 (Search, Learning, Information and Knowledge).

Executive Summary

This paper serves three functions:

1. Define Zero Click Marketing as a measurement problem grounded in cognitive psychology and brand-effectiveness research.
2. Review the empirical evidence base for distributed influence across search, social platforms, generative AI answer engines, and private channels.
3. Integrate Google Marketing Live 2026 as platform-owner evidence consistent with the directional thesis and as an inflection point in the measurement infrastructure that succeeds click attribution.

The dominant click-centric model of digital influence is no longer adequate when used as the sole measurement framework. The directional finding that a growing share of marketing influence appears to be formed before, outside, or without a click is supported by independent data (Pew Research Center 2025; Reuters Institute Digital News Report 2025), industry data (SparkToro/Datos, Ahrefs, Similarweb, Seer, Amisive, SISTRIX, BrightEdge, Bain & Co.), and now by evidence consistent with the directional thesis from the platform owner's own public position (Google Marketing Live 2026). The convergence of these three categories of evidence on the same directional claim, with Google having the strongest commercial incentive to dispute it, places the directional thesis in the strong-evidence tier. Magnitudes still vary across studies and methodologies.

The practical conclusion is straightforward: maintain clicks as one signal among several; instrument self-reported attribution, MMM, geo-experiments, and brand tracking; invest in distinctive, citable, native content that survives in LLM answer surfaces and in private-channel conversations. A new pragmatic note from GML 2026: the platform-owned successor metrics to click attribution (ABS, QFC) are powerful and granular but live entirely inside Google's measurement world. Triangulation with independent measurement remains necessary.

Figure 1. The Distributed Influence Measurement Problem

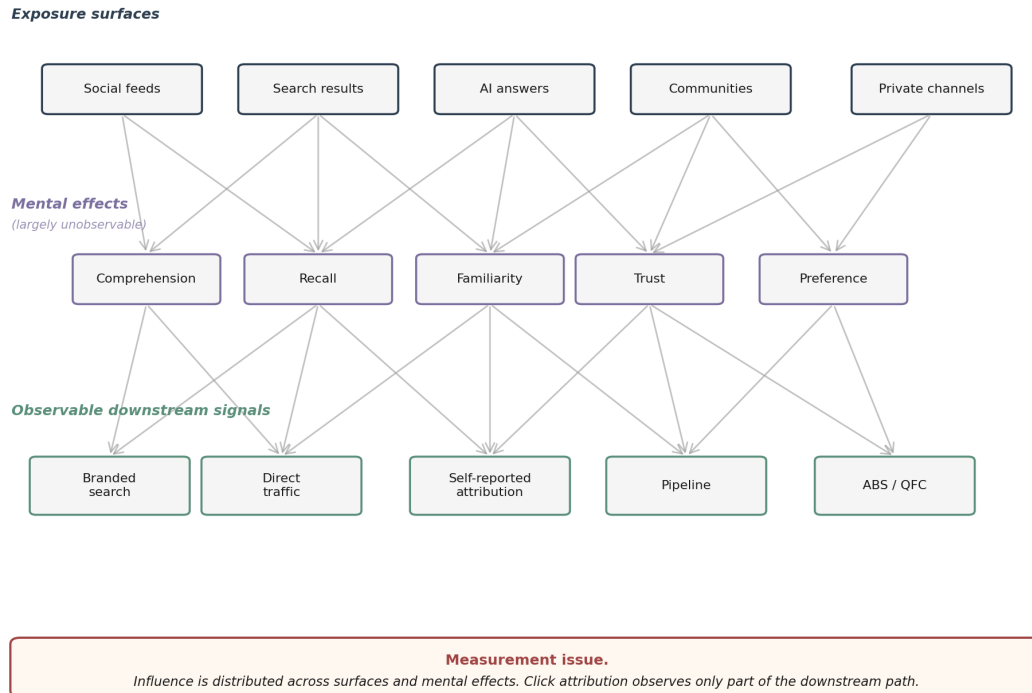


Figure 1: The Distributed Influence Measurement Problem. Exposure to a brand can occur across many surfaces (social feeds, search results, AI answers, communities, private channels). These surfaces drive largely unobservable mental effects (comprehension, recall, familiarity, trust, preference). The mental effects manifest in observable downstream signals (branded search, direct traffic, self-reported attribution, pipeline, ABS/QFC). Click attribution observes only part of this distributed path, which is the central measurement problem this paper addresses.

Section 1: Definition and Conceptual Boundaries

1.1 Operational definition

“Zero Click Marketing” denotes a creation, distribution, and measurement approach aimed at producing comprehension, recall, trust, preference, and future demand in environments where the audience can be influenced without visiting a brand’s owned property. The term was popularized by Rand Fishkin (SparkToro) beginning with his 2019 and 2020 “zero-click search” studies and his subsequent advocacy for “zero-click content.”

What it is: a strategy and measurement framework for environments where influence often occurs before, outside of, or without a click; the click can be a consequence rather than a precondition of conversion.

What it is not: an anti-click doctrine, a rejection of paid search or performance marketing, or a claim that clicks no longer matter. It is a critique of click-only attribution and click-bait-style distribution.

1.2 Boundary distinctions

- *Zero-click search:* SERP-resolved queries (featured snippets, knowledge panels, AI Overviews, Local Pack, calculators).
- *Zero-click social:* native posts that deliver complete value in-feed without an outbound URL.
- *Dark social:* traffic, conversations, and conversions propagated via private channels (DMs, Slack, WhatsApp, email forwards), commonly misclassified as “direct” in analytics. Documented by SparkToro (Fishkin 2020) and Refine Labs (Walker 2022–2024).
- *AI-mediated discovery:* surfaces (Google AI Overviews, Google AI Mode, ChatGPT, Perplexity, Gemini, Claude) where the model synthesizes an answer that may or may not cite sources.
- *Native content:* content optimized for the medium rather than as a teaser for an outbound link.
- *Ungated/gated content:* ungated content is publicly accessible and crawlable; gated content sits behind a form or login wall.
- *Measurement without click:* brand-lift studies, MMM, geo-experiments, self-reported attribution, branded-search lift, qualitative receipts.

1.3 Origin of the term

Fishkin’s “zero-click search” framing first reached mainstream marketing in his Jumpshot-based 2019 study; subsequent SparkToro/Datos panels (2020, 2021, 2024) extended the analysis. The broader “Zero Click Marketing” framing, extending from search behavior to all environments of pre-click influence, is industry-led but draws on robust academic foundations (Sharp’s mental availability; Binet & Field’s brand-vs-activation; Romaniuk’s CEPs).

1.4 GML 2026 reframing of the term

At Google Marketing Live 2026, Vidhya Srinivasan (VP/GM Search Ads) articulated what is effectively a platform-side acknowledgment of the zero-click paradigm: *“It’s time to acknowledge a fact. Google Search is AI Search through and through”* (~32:34). This is significant because the original “zero-click search” framing was a critique of Google by external researchers; Google has now publicly adopted parts of the underlying premise through its AI Search and measurement language, while restructuring major product surfaces accordingly. The boundary between “zero-click as external critique” and “zero-click as platform-recognized measurement problem” became substantially less clear in May 2026.

Section 2: Empirical Evidence on Click Decline and Transformation

2.1 Headline numbers across sources

A simplified summary table appears below. The full evidence matrix with sample sizes, methods, and detailed findings is provided in Appendix A.

Source	Year	Direction	Evidence type	COI flag
SparkToro/Datos (Fishkin)	2024	58.5% US / 59.7% EU Google searches end without click	Multi-million-device clickstream panel	Vendor (Datos is a Semrush company)
Pew Research Center	Jul 2025	AI Overviews roughly halve outbound clicks (8% vs 15%)	Browser-tracked KnowledgePanel	Independent non-profit (Google disputed methodology)
Ahrefs	Apr 2025 → Dec 2025	Position-1 CTR drop of 34.5% deepened to 58% in AIO-present SERPs	GSC + tool data, 300K keywords	Vendor (sells SEO tools)
Similarweb	Jul 2025	Zero-click news searches rose from 56% to ~69% in one year	Clickstream	Vendor (sells analytics)

Source	Year	Direction	Evidence type	COI flag
Amsive	2025	Average CTR drop 15.49%; non-branded -19.98%; branded +18.68%	GSC analysis, 700K keywords	Agency
Seer Interactive	2025	61% organic CTR decline; AIO citation lifts paid clicks 91%	Aggregated 42-client data	Agency
SISTRIX	2025	Position-1 CTR on AIO-present SERPs fell 27% → 11%	National panel, 100M+ German keywords	Vendor (sells SEO tools)
BrightEdge	May 2025	CTR fell 30% YoY while impressions rose 49%	Aggregated tracking	Vendor (sells SEO tools)
Bain & Company	Feb 2025	80% of consumers rely on AI for ≥40% of searches	Consumer survey	Management consultancy
Reuters Institute	2025	Social/video news 52% (2020) → 65% (2025)	Survey, 48 countries	Independent (Oxford)
Google (Srinivasan, GML 2026)	May 2026	AI Mode: 1B+ MAU, doubling each quarter; AI Overviews: 2.5B MAU	Internal Google data	Platform owner (direct commercial interest)
Google (Rettke, GML 2026)	May 2026	Only 40% of Demand Gen conversions occur in first 30 days	Internal Google data	Platform owner (direct commercial interest)

The convergence of independent, industry, and platform-owner evidence on the directional claim, with Google holding the strongest commercial incentive to dispute it, is the central observation supporting the strong-evidence tier classification.

2.2 Google's platform-owner evidence consistent with the directional thesis (GML 2026)

The most consequential evidence shift between the May 2025 and May 2026 versions of this investigation is the entry of Google itself as an evidence source. Until GML 2026, Google's official position consistently disputed external CTR studies. Pew Research Center's July 2025 study was officially contested by a Google spokesperson as using *"a flawed methodology and skewed queryset that is not representative of Search traffic."*

At Google Marketing Live 2026 (broadcast from Google's Bay View campus, May 20-21, 2026), Google's own product leadership made several statements that constitute platform-owner evidence consistent with the directional claim:

On AI Search adoption, Vidhya Srinivasan (VP/GM Search Ads) made five relevant claims:

1. *"AI Overviews... now has more than 2.5 billion monthly active users."* (~28:24)
2. *"AI Mode... has already surpassed more than a billion monthly users, and we're seeing phenomenal growth. Searches in AI mode have been doubling every quarter since launch."* (~29:04)
3. *"AI mode searches on average three times as long as traditional searches."* (~32:03)
4. *"Brainstorming searches are growing 30% faster than AI Mode searches overall."* (~32:30)
5. *"It's time to acknowledge a fact. Google Search is AI Search through and through."* (~32:34)

On the shift away from manually selected keywords, Srinivasan stated:

1. *"This means you can't think in terms of simple keywords anymore and you definitely can't manually choose them. And why would you? Just let the best AI do it for you."* (~33:00)
2. *"You can't choose keywords anymore."* (~54:59)

On ads as answers, Srinivasan stated:

1. *"Now you can ask Google anything, so the best ads must be answers."* (~50:51)
2. *"In all of these new formats, one thing is clear: Your brand doesn't just show up; it actually answers a question."* (~54:45)

On click-attribution as legacy, Nicky Rettke (VP YouTube Ads) stated:

1. *"Higher long-term ROAS is how you build a business that lasts, the difference between buying the click and investing in real growth."* (~1:20:25)
2. *"Finally, providing modern multi-touch attribution that goes beyond this legacy concept of just clicks."* (~1:20:18)
3. *"Only 40% of Demand Gen conversions happen in the first 30 days but really that's just the beginning."* (~1:21:31)

2.3 Evidence tier assessment

Strong evidence. The directional claim (AI Overviews and AI Mode reduce outbound clicks for informational queries; AI Search adoption is substantial and growing) is now corroborated by:

- Independent academic-adjacent research (Pew, Reuters Institute)
- Industry studies (Ahrefs, Similarweb, SparkToro/Datos, Seer, Amsive, SISTRIX, BrightEdge)
- **Platform-owner evidence consistent with the directional thesis (Google Marketing Live 2026)**

The convergence of these three categories, with Google having the strongest commercial incentive to dispute the narrative, places the directional thesis in the strong-evidence tier. This is an upgrade from the May 2025 version of this investigation, which placed it in moderate-strong.

Moderate evidence. Specific magnitude claims (Pew’s 46.7% relative click reduction, Ahrefs’ 58% position-1 drop, Similarweb’s 56%→69% news zero-click) remain methodologically heterogeneous. The directional pattern is well-supported; specific magnitudes remain study-dependent.

Weak / anecdotal. Industry blog claims about “the end of search” rest on extrapolation. Same-keyword Semrush analysis showed only 38.1%→36.2% zero-click change pre/post AIO; the larger movement is driven by AIO-eligible queries growing as a share, not by within-query CTR collapse.

2.4 ChatGPT referrals

Similarweb (July 2025) reports ChatGPT referrals to news sites grew approximately 25× year-over-year (from under 1M visits Jan-May 2024 to over 25M Jan-May 2025); news-related prompts in ChatGPT rose 212% over the same period. This signal is corroborated by Reuters Institute survey data showing 54% of respondents saw an AI-generated search answer in the past week.

Section 3: Social Platforms and Native Content

3.1 LinkedIn

LinkedIn Engineering’s feed-ranking architecture (First Pass Rankers + Second Pass Ranker + reranker) **explicitly downweights binary click signals and upweights dwell time**: *“Click and viral actions can be rare, especially for passive consumers of the feed... primarily binary indicators of engagement”* (LinkedIn Engineering, “Understanding Feed Dwell Time”). This algorithmic design tends to favor content consumed on-platform over content that asks users to click out.

Hootsuite controlled experiment (Jan 22 - Mar 22, 2021): 177 link posts vs 7 linkless across Hootsuite’s own LinkedIn account. *“On average, posts without links*

got 6× more reach than posts with links... almost 4× more reactions and 18× more comments.”

Richard van der Blom’s Algorithm Insights Reports (2024-2026): based on ~1.3M LinkedIn posts, *“One external link in the post body reduces median reach by 18.8%... LinkedIn now suppresses comments containing external links, with visibility reduced by up to 80%.”*

Hootsuite’s official 2025 guidance: *“LinkedIn is pushing more native content, like text posts, carousels, and videos, over posts with outbound links. If you need to share a link, consider placing it in the comments instead.”*

3.2 X (Twitter)

Elon Musk publicly confirmed (Oct 2023 on X): *“Our algorithm tries to optimize time spent on X, so links don’t get as much attention, because there is less time spent if people click away. Best thing is to post content in long form on this platform.”*

Twitter’s open-sourced algorithm (Mar 2023, refreshed Jan 2026 with Grok integration) confirms engagement weights (a reply $\approx$ 27 retweets; bookmark multiplier $\times 10$ vs $\times 0.5$ for likes). Independent A/B tests report 50–94% reach reductions for posts with external links versus identical content without; magnitudes are variable across creators.

3.3 Meta (Facebook, Instagram, Threads)

Adam Mosseri (head of Threads/Instagram): *“We don’t downrank links, but we don’t place much value on p(click) and people don’t like and comment on links much.”* This creates a practical distribution disadvantage without requiring an explicit link penalty.

Recent Meta Widely Viewed Content reports consistently show that the overwhelming majority of U.S. Facebook Feed views are not views of posts with external links.

3.4 TikTok

TikTok’s official documentation identifies three ranking inputs: *“User interactions, Content information, User information.”* External links are not listed as a ranking factor. Bio link clicks are a separate UX layer with much smaller distribution.

3.5 Synthesis

Across major platforms there is **moderate-to-strong evidence** that posts which keep users on-platform are algorithmically advantaged either explicitly (X) or through dwell-time and engagement weighting (LinkedIn, Meta). TikTok’s public documentation does not identify external links as a ranking input.

Conflict of interest flag: Buffer, Hootsuite, Metricool, Sprout Social, and Analytica all sell scheduling/listening tools. Van der Blom’s consulting practice depends on the dynamics he studies.

Section 4: Psychology and Cognitive Science of Zero Click Marketing

4.0 Why this section matters

The “Zero Click Marketing” narrative often rests on intuitions about how people perceive, remember, and decide. Industry advocates routinely invoke mere exposure, processing fluency, mental availability, and the 95-5 rule as if these were settled, unified mechanisms operating predictably in social feeds and AI answer surfaces. They are not. Each construct has its own empirical literature, with its own boundary conditions, effect sizes, and replication record. This section consolidates what the peer-reviewed evidence actually says, including the limits and confounds the popular narrative tends to skip.

The purpose of this section is not to prove that zero-click exposure always creates demand. It is to identify which psychological mechanisms make the hypothesis plausible and where the causal chain remains untested.

Five questions structure the section: 1. Does repeated non-clicked exposure to a brand affect liking and recognition? (Mere exposure) 2. Does easier processing produce more favorable evaluation? (Processing fluency) 3. Does exposure encode into memory without conscious attention? (Implicit and low-attention processing) 4. Does the timing of exposures matter? (Distributed practice and spacing) 5. Do these effects survive in real digital-feed contexts? (Applied evidence)

A sixth question, which the literature does not yet answer well, closes the section: *does the chain Exposure → Comprehension → Memory → Familiarity → Trust → Preference → Search hold end-to-end for zero-click content?*

4.1 Mere Exposure: from Zajonc to Montoya

The foundational claim

Robert Zajonc’s monograph “Attitudinal effects of mere exposure” (1968, *Journal of Personality and Social Psychology Monograph Supplement* 9(2, Pt. 2), 1-27; DOI: 10.1037/h0025848) established the empirical observation that “*mere repeated exposure of the individual to a stimulus is a sufficient condition for the enhancement of his attitude toward it.*”

The first meta-analysis

Robert Bornstein’s 1989 meta-analysis (*Psychological Bulletin* 106(2), 265-289; DOI: 10.1037/0033-2909.106.2.265) synthesized 208 experiments. Key findings:

- Average effect size $r \approx 0.26$ (small-to-medium)
- Effects were stronger for unfamiliar stimuli than for familiar ones
- Effects were stronger for brief exposures than long ones

- Effects were stronger when exposure was subliminal (below conscious awareness) than supraliminal

The subliminal finding is counterintuitive but replicated: **Bornstein & D’Agostino (1992)** (*Journal of Personality and Social Psychology* 63(4), 545-552) demonstrated that 5-millisecond stimuli produced *larger* mere-exposure effects than 500-millisecond stimuli. This is the empirical basis for the claim that scroll-past content can affect preference even when participants cannot consciously recall having seen it.

Limits and overexposure

Bornstein, Kale & Cornell (1990) “Boredom as a limiting condition on the mere exposure effect” (*Journal of Personality and Social Psychology* 58(5), 791-800; DOI: 10.1037/0022-3514.58.5.791) documented that the effect degrades with overexposure. After a sufficient number of repetitions, additional exposures reduce liking. This is the wear-out boundary that Zero Click Marketing advocates rarely acknowledge.

The 2017 re-examination

Montoya, Horton, Vevea, Citkowicz & Lauber (2017) “A re-examination of the mere exposure effect: The influence of repeated exposure on recognition, familiarity, and liking” (*Psychological Bulletin* 143(5), 459-498; DOI: 10.1037/bul0000085) is the most recent comprehensive synthesis, building on but going beyond Bornstein 1989. Methodologically, the authors used latent growth curve modeling, which Bornstein could not, allowing them to model nonlinear dose-response patterns.

Key findings:

- 268 curve estimates from 81 articles
- The mere exposure effect is characterized by a **positive slope and negative quadratic effect**, i.e., an **inverted-U shape**: liking increases with exposure to a peak, then declines
- The effect was larger for **ideographs (slope = 336.63)**, nonsense words (slope = 291.23), and **paintings/drawings/matrices (slope = 264.38)** than for **photographs (slope smaller)**
- A causal-mediation analysis suggested that **recognition mediates liking**, not the other way around. This reverses an older assumption that liking might cause recognition. It supports the operational claim that getting consumers to *recognize* a brand from a distinctive asset is the foundational step that enables liking.

Implications for Zero Click Marketing: 1. The effect is real, replicated across nearly 60 years and two major meta-analyses. 2. The effect saturates and reverses. **There is no “more is always better” with brand exposure.** This contradicts naive “frequency at all costs” thinking common in always-on social. 3. The effect is **larger for novel/distinctive stimuli** than for ordinary photographs. This supports Romaniuk’s Distinctive Brand Assets work: stylized, ownable visual or verbal devices out-

perform generic imagery. 4. The fact that **recognition mediates liking** anchors the “Remembered” rung of the Zero Click Ladder as causally upstream of the “Preferred” rung. This is not idle structure; it is testable.

4.2 Processing Fluency

The construct

Processing fluency is the subjective ease with which information is encoded. Easier-to-process information feels more familiar, more trustworthy, more pleasant.

Foundational papers

Reber, Winkielman & Schwarz (1998) “Effects of perceptual fluency on affective judgments” (*Psychological Science* 9(1), 45-48; DOI: 10.1111/1467-9280.00008): primed/easier-to-process stimuli were rated more favorably than equivalent harder-to-process stimuli.

Reber, Schwarz & Winkielman (2004) “Processing fluency and aesthetic pleasure: Is beauty in the perceiver’s processing experience?” (*Personality and Social Psychology Review* 8(4), 364-382; DOI: 10.1207/s15327957pspr0804_3) is the canonical theoretical review. The core claim: “*The more fluently perceivers can process an object, the more positive their aesthetic response.*”

The broader review

Alter & Oppenheimer (2009) “Uniting the tribes of fluency to form a metacognitive nation” (*Personality and Social Psychology Review* 13(3), 219-235; DOI: 10.1177/1088868309341564) integrated the fluency literature across linguistic, perceptual, conceptual, and decision domains. Fluency influences judgments of truth, frequency, confidence, familiarity, liking, and risk.

Schwarz (2004) “Metacognitive experiences in consumer judgment and decision making” (*Journal of Consumer Psychology* 14(4), 332-348; DOI: 10.1207/s15327663jcp1404_2) translated fluency into consumer-behavior terms.

Implication for Zero Click Marketing

Content optimized for the platform it lives on (native layout, no broken images, no consent banners, no off-platform jumps) is processed more fluently than content built around a click-out. The fluency literature predicts that, holding the underlying message constant, native delivery produces more favorable evaluation and easier recall.

Caveat. Fluency effects are typically small to moderate. Over-fluency can backfire when readers attribute fluency to manipulation (Briñol, Petty & Tormala 2006 *Journal of Consumer Research*).

4.3 Distributed Practice and the Spacing Effect

Why this matters for Zero Click Marketing

Most “Zero Click” content programs ask: “*how many exposures and on what schedule?*” The answer lies in the spacing-effect literature, which is one of the most-replicated findings in cognitive psychology, going back to Ebbinghaus (1885). The empirical question is: does massed (clustered) or distributed (spaced) repetition produce better retention?

The canonical meta-analysis

Cepeda, Pashler, Vul, Wixted & Rohrer (2006) “Distributed practice in verbal recall tasks: A review and quantitative synthesis” (*Psychological Bulletin* 132(3), 354-380; DOI: 10.1037/0033-2909.132.3.354) is the canonical synthesis: 839 assessments of distributed practice in 317 experiments across 184 articles.

Key finding: “*When retention intervals are measured at a week or greater, studies show that distributed practice nearly doubles recall performance*” (Cepeda et al. 2006). The effect of spacing is large, not small.

The temporal ridgeline

Cepeda, Vul, Rohrer, Wixted & Pashler (2008) “Spacing effects in learning: A temporal ridgeline of optimal retention” (*Psychological Science* 19(11), 1095-1102; DOI: 10.1111/j.1467-9280.2008.02209.x) tested over 1,350 individuals across delays of up to 1 year.

Key finding: “*The optimum gap value was about 20% of the test delay for delays of a few weeks, falling to about 5% when delay was one year.*”

This is the most operationally consequential finding in the zero-click psychology literature, because it directly translates to content frequency planning:

- If you want a customer to remember your brand **next week**: optimal gap between exposures $\approx$ 20% of the interval, roughly 1-2 day spacing
- If you want them to remember **in three months**: optimal gap $\approx$ 15-20 days
- If you want them to remember **in a year** (typical B2B 95-5 horizon): optimal gap $\approx$ 18 days at the start, expanding

Implications for Zero Click Marketing

1. Daily bombardment is unlikely to be optimal for long-cycle purchases. For B2B SaaS or medical-tourism category with 6-18 month consideration windows, exposures should be spaced roughly 1-3 weeks apart, not daily.
2. The spacing literature provides a more defensible starting point for frequency planning than vendor guidance alone.
3. The 95-5 rule (Dawes 2021) and the spacing effect literature are mathematically compatible: long-horizon retention requires longer inter-exposure intervals than short-horizon.

Important caveat on extrapolation. The Cepeda et al. (2006, 2008) findings were established in verbal-recall and educational paradigms (vocabulary, facts, conceptual learning), not in marketing-frequency contexts. The application of the temporal-ridgeline result to advertising frequency and brand-exposure spacing is a working hypothesis defended on mechanistic grounds (both contexts involve repeated cued retrieval from long-term memory), not a demonstrated empirical finding for the marketing case. The translation should be treated as theoretically motivated and operationally useful pending direct empirical replication in feed-exposure or display-advertising paradigms.

4.4 Implicit Memory and Low-Attention Processing

The construct

Most ads and social posts are not consciously attended to. Yet the brain processes them, and they leave traces. The literature on **implicit memory** is the relevant body of evidence.

Foundational work

Krishnan & Trappey (1999) “Nonconscious memory processes in marketing: A historical perspective and future directions” (*Psychology and Marketing* 16(6), 451-457) reviewed the marketing-side application of implicit memory.

Janiszewski (1993) “Preattentive mere exposure effects” (*Journal of Consumer Research* 20(3), 376-392) demonstrated that brand exposure in peripheral vision (not foveated) still produced mere exposure effects.

The Low Attention Processing theory

Heath (2001) *The Hidden Power of Advertising: How Low Involvement Processing Influences the Way We Choose Brands* (Admap Monographs 7) and the academic articles surrounding it argued that most advertising effects occur below the threshold of conscious attention. **Heath (2012)** *Seducing the Subconscious: The Psychology of Emotional Influence in Advertising* (Wiley-Blackwell) extended the argument with neuroscience evidence.

Heath’s central claim is empirically defensible: peripheral and low-attention exposures build associative memory networks that influence later choice without entering explicit recall. **This is the mechanism by which scrolled-past content can still affect preference.** It is also the mechanism the Pew Research Center 2025 study indirectly captures when AI Overviews produce branded-search lift without click-through.

Karen Nelson-Field’s attention research

The Amplified Intelligence work (multiple industry reports) connects directly to this literature. Nelson-Field documents that “on average, only about 20% of impressions are actively seen, while 80% are wasted,” and “85% of digital impressions last under

2.5 seconds,” which is below conventional thresholds for explicit memory formation. The 2025 **VCCP Media / Nelson-Field** *Hacking the Attention Economy* report estimated that 1.5 seconds of active attention with distinctive brand assets is sufficient to encode memory in real-world digital feeds.

Caveat (industry source flag). Nelson-Field’s work, while methodologically careful, is published in industry venues and consulting reports, not peer-reviewed journals. The numbers should be treated as practitioner evidence pending academic replication.

4.5 Limited Capacity Theory of Mediated Message Processing

Annie Lang (2000) “The limited capacity model of mediated message processing” (*Journal of Communication* 50(1), 46-70; DOI: 10.1111/j.1460-2466.2000.tb02833.x) and its 2006 extension propose a finite cognitive-resource model with three sub-processes: encoding, storage, retrieval.

Implication for Zero Click Marketing. Teaser-with-link content requires the user to allocate cognitive resources both to processing the message AND to making the click decision. Native complete content allocates all resources to the message. Lang’s model predicts that, for equivalent message content, native delivery produces deeper encoding and better storage.

4.6 Levels of Processing

Craik & Lockhart (1972) “Levels of processing: A framework for memory research” (*Journal of Verbal Learning and Verbal Behavior* 11(6), 671-684) established that deeper semantic processing produces better recall than shallow perceptual processing.

Craik (2002) “Levels of processing: Past, present... and future?” (*Memory* 10(5-6), 305-318) is the modern review.

Implication for Zero Click Marketing. Complete native content with substantive argumentation engages semantic processing; teaser content engages shallower perceptual or surface processing. Levels-of-processing theory predicts better recall for complete native content holding exposure time roughly constant.

4.7 Visual Attention in Advertising (eye-tracking evidence)

The Pieters & Wedel research program, conducted at Tilburg University and the University of Michigan, is the most rigorous body of eye-tracking evidence on advertising attention.

Pieters, Rosbergen & Wedel (1999) “Visual attention to repeated print advertising: A test of scanpath theory” (*Journal of Marketing Research* 36(4), 424-438; DOI: 10.1177/002224379903600403): attention duration decreases significantly across repetitions; attentional scanpaths remain stable.

Pieters & Wedel (2004) “Attention capture and transfer in advertising: Brand, pictorial, and text-size effects” (*Journal of Marketing* 68(2), 36-50; DOI: 10.1509/jmkg.68.2.36.27794):

based on 1,363 print advertisements tested with infrared eye-tracking on more than 3,600 consumers. Three findings:

- The pictorial captures attention independent of its size
- Text captures attention in direct proportion to its surface size
- **The brand element most effectively transfers attention to the other elements**

Wedel & Pieters (2008) “Eye tracking for visual marketing” (*Foundations and Trends in Marketing* 1(4), 231-320; DOI: 10.1561/1700000011) is the comprehensive review.

Implication for Zero Click Marketing. Visual elements in native content are not interchangeable. The brand element (logo, signature, distinctive verbal device) transfers attention to other elements; without it, attention disperses. This empirically supports Romaniuk’s Distinctive Brand Assets framework.

4.8 Consumer Neuroscience

The field

Plassmann, Ramsøy & Milosavljevic (2012) “Branding the brain: A critical review and outlook” (*Journal of Consumer Psychology* 22(1), 18-36; DOI: 10.1016/j.jcps.2011.11.010) is the foundational critical review. It catalogs both the promises and the limits of fMRI/EEG/biometric methods in marketing research.

Plassmann, Venkatraman, Huettel & Yoon (2015) “Consumer neuroscience: Applications, challenges, and possible solutions” (*Journal of Marketing Research* 52(4), 427-435; DOI: 10.1509/jmr.14.0048) is the practitioner-facing review.

Specific findings relevant to brand exposure

McClure, Li, Tomlin, Cypert, Montague & Montague (2004) “Neural correlates of behavioral preference for culturally familiar drinks” (*Neuron* 44(2), 379-387; DOI: 10.1016/j.neuron.2004.09.019) is the famous Coke vs. Pepsi fMRI study. Brand label exposure activated medial prefrontal cortex and hippocampus, regions associated with cultural information and emotional memory. In blind tests, preferences did not differ; in branded tests, they did, with corresponding neural activity differences.

Plassmann, O’Doherty, Shiv & Rangel (2008) “Marketing actions can modulate neural representations of experienced pleasantness” (*PNAS* 105(3), 1050-1054; DOI: 10.1073/pnas.0706929105) demonstrated that price labels altered medial orbitofrontal cortex activity for identical wines.

Berns & Moore (2012) “A neural predictor of cultural popularity” (*Journal of Consumer Psychology* 22(1), 154-160) showed that adolescent fMRI responses to songs predicted commercial success three years later, even though self-report responses did not.

Implication for Zero Click Marketing

Brand exposure produces measurable neural signatures even without behavioral preference change. The “Remembered” and “Preferred” rungs of the Zero Click Ladder have neural correlates that explicit-measure tracking does not capture.

Caveat. Consumer neuroscience is an expensive, low-volume measurement methodology. Its value here is theoretical: it provides converging evidence that non-clicked exposure produces real cognitive change.

4.9 Mental Availability, Distinctive Assets, and Category Entry Points

Mental availability (Sharp & Romaniuk)

Sharp (2010) *How Brands Grow* (Oxford UP) and **Romaniuk & Sharp (2016)** *How Brands Grow Part 2* (Oxford UP) argue brand growth is driven primarily by **mental availability** (propensity to be retrieved in buying situations) and **physical availability** (propensity to be encountered). Mental availability is operationally a recognition-and-retrieval construct, directly downstream of the mere-exposure and processing-fluency mechanisms documented above.

Distinctive Brand Assets

Romaniuk (2018) *Building Distinctive Brand Assets* (Oxford UP) defines distinctive assets as non-brand-name elements (colors, characters, taglines, typography, sounds) that consumers use to identify a brand. The “Distinctive Assets Grid” measures assets along two axes:

- **Fame:** % of category buyers who link the asset to any brand
- **Uniqueness:** % of those who link it specifically to the target brand

Romaniuk’s empirical work shows that brands with strong distinctive assets sustain mental availability at lower media weights. The Montoya et al. (2017) finding that ideographs and stylized visual stimuli produce stronger mere-exposure effects than ordinary photographs is the cognitive-science underpinning of this practitioner framework.

Category Entry Points

Romaniuk’s 7W framework for Category Entry Points (Why, When, Where, While, With/For Whom, With What, hoW-feeling) operationalizes mental availability as a network of associative cues linking brand to buying occasions. CEPs are the consumer-research-tradition counterpart of associative-memory network theory (Anderson 1983 *The Architecture of Cognition*).

4.10 Long-Term Effects and the 95-5 Rule

Long and Short of It

Binet & Field (2013) *The Long and the Short of It* (IPA) and **Binet & Field (2018)** *Effectiveness in Context* (IPA) synthesized 996 IPA Effectiveness Award cases over 30+ years. The 60:40 brand-to-activation ratio is the most-replicated finding in marketing-effectiveness research.

This is convergent with **Cepeda et al. (2008)** on the temporal ridgeline: longer-horizon retention requires sustained, spaced exposure (the “long” brand-building investment), not concentrated activation pulses. Two methodologically independent literatures arrive at compatible conclusions about temporal allocation.

95-5 Rule

Dawes (2021), with LinkedIn B2B Institute, formalized the empirical observation that only ~5% of B2B buyers are in-market at any given quarter. The implication for measurement: 95% of valuable advertising effect is *future-period* recall building, mediated by mental availability and the mere-exposure / spacing-effect mechanisms above.

The 95-5 rule is consistent with the Cepeda 2008 temporal-ridgeline finding: for 1-year retention horizons, exposures should be spaced at roughly 15-20% of the retention interval, meaning weekly-to-bi-monthly exposure for B2B categories with 6-18 month buying cycles.

4.11 Source Credibility, Persuasion, and Trust Transfer

Foundational

Hovland, Janis & Kelley (1953) *Communication and Persuasion* (Yale UP) established source credibility (expertise + trustworthiness) as a primary persuasion variable.

Petty & Cacioppo (1986) Elaboration Likelihood Model (*Advances in Experimental Social Psychology* 19, 123-205; DOI: 10.1016/S0065-2601(08)60214-2) introduced the peripheral-route vs. central-route distinction. In zero-click contexts where engagement is shallow, peripheral cues (source expertise, social proof, distinctive assets) dominate; in deep-attention contexts, message arguments dominate.

Implicit attitudes

Greenwald & Banaji (1995) “Implicit social cognition: Attitudes, self-esteem, and stereotypes” (*Psychological Review* 102(1), 4-27; DOI: 10.1037/0033-295X.102.1.4) established the framework for implicit attitudes. The Implicit Association Test (Greenwald, McGhee & Schwartz 1998) has been applied to brand attitudes in numerous subsequent studies, demonstrating that brand attitudes formed through repeated exposure show up in implicit measures even when explicit attitudes are unchanged.

Trust transfer

Stewart (2003) “Trust transfer on the World Wide Web” (*Organization Science* 14(1), 5-17) documented that trust in one entity (a recommender, a community, a creator) transfers to recommended entities. This is the empirical anchor for influencer marketing, podcast-host endorsements, and Reddit-thread effects.

Implication for Zero Click Marketing

A brand mentioned by a trusted creator on a platform inherits credibility without the user clicking through to the brand. The trust-transfer mechanism is one of the cleanest explanations for why dark-social (DM, podcast, Reddit) channels drive disproportionate self-reported attribution despite being invisible to click-attribution.

4.12 Negativity Bias and Emotional Encoding

Baumeister, Bratslavsky, Finkenauer & Vohs (2001) “Bad is stronger than good” (*Review of General Psychology* 5(4), 323-370; DOI: 10.1037/1089-2680.5.4.323) synthesized evidence that negative information is weighted more strongly than positive in attention, memory, and behavior.

Rozin & Royzman (2001) “Negativity bias, negativity dominance, and contagion” (*Personality and Social Psychology Review* 5(4), 296-320) developed the theoretical framework.

Implication for Zero Click Marketing. Negative brand-relevant content (a critical review on Reddit, a complaint on X, a negative comparison in a YouTube video) has disproportionate impact relative to its frequency. The corollary: brand presence in answer-engine responses where a critical review or negative framing is featured does not produce neutral exposure; it produces net-negative attitude shift. This is a counterweight to the simple “all exposure helps” reading of mere-exposure findings.

4.13 The Limited and the Promising: Caution on the Causal Chain

The proposed Zero Click Ladder is:

Distributed exposure → comprehension → memory → familiarity → trust → preference → active search / conversion / recommendation

Each link has supporting peer-reviewed evidence:

Link	Anchoring evidence	Status
Exposure → comprehension	Lang 2000 limited capacity; Craik & Lockhart 1972 levels of processing	Supported
Comprehension → memory	Craik 2002; semantic-deep-processing literature	Supported

Link	Anchoring evidence	Status
Exposure → memory (implicit)	Bornstein 1989; Montoya et al. 2017; Janiszewski 1993; Heath 2001	Strong
Memory → familiarity	Mandler 1980 recognition memory; Montoya et al. 2017 recognition mediates liking	Strong
Familiarity → liking	Reber et al. 1998, 2004; Bornstein 1989	Strong
Liking → preference	Aaker 1991; Keller 1993; consumer-behavior tradition	Moderate (preference has many confounds)
Preference → search behavior	Less direct empirical work; Romaniuk Better Brand Health 2023 connects	Weak-Moderate

The full chain has not been tested end-to-end in a single zero-click context. Each link is supported by its own peer-reviewed evidence base, but the integration is the contribution that Studies 2 and 3 of the proposed research protocol would establish. The honest scientific stance is that the Zero Click Ladder is **a plausible working model with empirical support for each link**, not a demonstrated causal pipeline.

Specific cautions

1. **Overexposure reverses the mere exposure effect** (Bornstein, Kale & Cornell 1990; Montoya et al. 2017 inverted-U). Continuous frequency is not free.
2. **Recognition does not equal preference** (Aaker 1991). The mediation in Montoya et al. 2017 runs through recognition but is not unconditional.
3. **Fluency can backfire** when readers attribute it to manipulation (Briñol, Petty & Tormala 2006).
4. **Spacing is non-monotonic** (Cepeda et al. 2008). Daily and yearly are both wrong; the right interval depends on the retention horizon.
5. **Brand effects vary by category and buyer state** (Binet & Field; Dawes 95-5). FMCG and high-consideration B2B operate on different temporal regimes.
6. **Single-exposure laboratory paradigms are weak ecological proxies** for multi-touch feed exposure.

4.14 GML 2026 Convergence with Binet & Field

The most significant theoretical convergence at Google Marketing Live 2026 was Nicky Rettke’s framing of long-term ROAS, which articulates Binet & Field’s 60:40 brand-to-activation ratio from the measurement side:

“Higher long-term ROAS is how you build a business that lasts, the difference between buying the click and investing in real growth.” (~1:20:25)

Google’s quantification of the temporal split: *“Only 40% of Demand Gen conversions happen in the first 30 days.”*

This is effectively the 60:40 ratio inverted to its measurement-side counterpart: 60% of conversion value emerges from brand-building “long” effects measurable only at horizons beyond 30 days; 40% from activation “short” effects measurable in the standard attribution window. The convergence of:

- Independent IPA effectiveness research (Binet & Field): 60:40 budget optimum
- Google’s product-side telemetry: 40% / 60% temporal split in conversions
- The Cepeda et al. (2008) temporal ridgeline: long-retention requires sustained spaced exposure
- The 95-5 rule (Dawes 2021): only 5% of buyers are in-market at any moment

on the same temporal structure is itself an evidence-strengthening triangulation across four methodologically independent traditions, even with Google’s COI noted.

Section 5: Operationalizing the Zero Click Ladder

The original ladder (Seen → Understood → Remembered → Preferred → Sought Out) proposed observable variables for each rung. Google Marketing Live 2026 announcements provide platform-level instrumentation for the bottom of the ladder. For readability, the original six-column ladder is split here into two tables: Table 2A presents the construct definitions and observable variables; Table 2B presents the measurement risks and the new Google-platform metrics introduced at GML 2026.

Table 2A. Zero Click Ladder: constructs and observable variables

Rung	Construct (academic anchor)	Observable variables	Original metric
Seen	Reach × frequency in target (Romaniuk 2023)	Audience-qualified impressions	Reach %, frequency, % in target
Understood	Comprehension (cognitive psych.)	Verbatim recall of key claim	Message-comprehension test
Remembered	Brand recall, recognition (Keller 1993; Romaniuk 2018)	Unaided/aided recall; brand-CEP association	% unaided recall; CEP linkage score
Preferred	Consideration, trust, preference (Hovland; ELM)	Consideration-set size; brand attitude	Net favorability; consideration; intent

Rung	Construct (academic anchor)	Observable variables	Original metric
Sought Out (short-term)	Active behavior	Branded search; direct traffic; demos; HDYHAU; referrals	Branded search index; direct/organic ratio; pipeline
Sought Out (long-term)	Sustained active behavior over months	Multi-month branded search lift; future conversion	Long-term ROAS

Table 2B. Zero Click Ladder: measurement risks and Google-platform metrics (GML 2026)

Rung	GML 2026 platform metric	Risk of misinterpretation
Seen	Standard platform metrics	Reach $\neq$ attention; bots; viewability fraud
Understood	Survey-only; not productized	Social-desirability bias; leading questions
Remembered	Survey-only; not productized	Recency bias; cued probes inflate scores
Preferred	Survey-only; not productized	Stated preference $\neq$ revealed preference
Sought Out (short-term)	Attributed Branded Searches (ABS)	Branded search has confounds (PR spikes, competitor outages)
Sought Out (long-term)	Qualified Future Conversions (QFC), 6-month horizon	Platform-owned; closed ecosystem

5.1 ABS: Attributed Branded Searches

Gaurav Bhaya (GM/VP Google Ads) at GML 2026: *“First, Attributed Branded Searches now available in Google Ads and Google Marketing Platform. This is the clear signal that your ad created intent.”* (~1:37:56)

ABS measures, within Google’s ecosystem, the lift in branded search behavior attributable to an ad exposure. It operationalizes the “Sought Out” rung of the Zero Click Ladder.

5.2 QFC: Qualified Future Conversions

Bhaya: *“This is why we are introducing a new powerful metric: Qualified Future Conversions. They connect signals such as follow-up branded searches, video views or site visits up to six months in advance.”* (~1:38:22)

QFC is Google’s productization of long-term ROAS measurement. Case example: UK retailer Crew Clothing’s YouTube prospecting campaign *“looked flat over a 30-day window... [under QFC] a 70% uplift in long-term conversions”* (~1:38:50).

5.3 Critical reading

The top three rungs (Understood, Remembered, Preferred) are better measured with survey instruments because comprehension, recall, and trust are mental states that platform behavior cannot fully proxy. The bottom two rungs are now Google-native metrics.

This resembles the measurement architecture the original investigation proposed; Google has now productized part of the lower-funnel measurement layer. The unresolved problem: ABS and QFC are visible only inside Google’s ecosystem. Non-Google-driven exposure that produces the same downstream behaviors will still be invisible to ABS-based attribution (it will appear in ABS as Google’s own work). The full Zero Click Ladder still requires a measurement stack that includes survey-based brand tracking + MMM + ABS/QFC, not ABS/QFC alone.

Section 6: Attribution vs Measurement

6.1 The distinction

Attribution distributes credit across observed touchpoints; **measurement** estimates causal lift. These are distinct activities frequently conflated.

6.2 Method comparison

Method	Use case	Failure modes	Primary citation
Last-click / First-click	Quick reporting	Massively overweights demand-capture channels; blind to dark social	Google Analytics docs
Linear / Time-decay / Position-based MTA	Cross-channel reporting	GIGO without cookies/IDs; over-claims digital	IAB / MMA Cross-Media Measurement Standards
Data-driven attribution (Google Ads/GA4)	DDA within Google ecosystem	Black-box; biased toward Google’s own surfaces	Google Ads Help

Method	Use case	Failure modes	Primary citation
Self-reported attribution (HDYHAU)	Capturing dark social	Recency and emotional-vividness bias; ~30% non-response if optional	Refine Labs; Fairing; SparkToro 2022 dark-social study
MMM: Meta Robyn, Google Meridian/LightweightMMM, PyMC-Marketing	Cross-channel budget allocation; privacy-safe	Confounded by trend/seasonality; requires 2+ years history; needs experiment calibration	Runge et al. 2024 arxiv:2403.14674
Geo experiments (GeoLift, GeoX)	Causal lift on geographic split	Spillover; few-geo problems	Vaver & Koehler 2011; Kerman, Wang, Vaver 2017
BSTS: CausalImpact	Quasi-experimental lift from a single intervention	Donor-market quality; structural breaks	Brodersen, Gallusser, Koehler, Remy & Scott (2015) <i>Annals of Applied Statistics</i> 9:247-274
Difference-in-differences Holdout / Conversion-lift	Policy/launch evaluations Causal lift on a specific channel	Parallel-trends assumption Sample size; ethical	Card & Krueger 1994 Facebook Conversion Lift docs
Qualitative receipts	Triangulating dark social	Cherry-picking	Refine Labs Hybrid Attribution Framework
ABS (Attributed Branded Searches)	Causal signal: ad created intent within Google ecosystem	Walled-garden capture; non-Google lift counted as Google	Google GML 2026 announcement
QFC (Qualified Future Conversions)	Long-term causal signal, 6-month horizon	Same walled-garden constraint	Google GML 2026 announcement

6.3 Refine Labs hybrid attribution data

A 12-month study covering 620 declared-intent conversions and \$21.5MM ARR reported a ~90% measurement gap between software-attribution and self-reported attribution; podcast was credited with 53% of revenue (\$11.4MM) by customers but ~0% by software. Chris Walker (LinkedIn, 2023): demand-capture attribution was 82% Google/Direct, 10% Social, 8% Other; demand-creation self-reported was 44%

Social, 30% Podcast, 13% Community, 10% Word of Mouth.

Conflict of interest. Refine Labs sells demand-creation consulting. The data is suggestive, not generalizable, and based on a SaaS B2B client base.

6.4 GML 2026: Google frames click-only attribution as a legacy model

At GML 2026, Nicky Rettke (in the announcement of TransUnion partnership) stated: *“Finally, providing modern multi-touch attribution that goes beyond this legacy concept of just clicks”* (~1:20:18). This is reflected in Google’s public product and measurement direction, made on its largest public stage.

Google’s announced measurement stack for 2026: 1. **First-party data integration** via Google Tag and Data Manager. 2. **Cross-channel MMM** via Meridian (open source) integrated into GA360. 3. **Causality signals**: ABS for short-term, QFC for long-term. 4. **Cross-channel reporting**: GA 360 now ingests TikTok, Pinterest, Snap impression and cost data. 5. **Agentic interface**: natural-language queries against the unified data model.

Bhaya: *“For the first time, you can analyze your performance across Google, TikTok, Pinterest, Snap and other social media platforms, all in one place”* (~1:39:35).

This broadly resembles the methodological architecture this investigation advocates. It is now reflected in Google’s public product and measurement direction, not only in academic measurement literature.

6.5 Critical reading

Meridian is open source. ABS and QFC are not. The choice of what to release matters: Meridian is a generic library that competitors and consultancies can use; ABS and QFC are tied to Google’s data pipelines and reporting. Marketers using only ABS/QFC for measurement risk adopting a Google-shaped view of attribution.

6.6 Evidence tier

Strong evidence. MMM, CausalImpact, and geo-experimentation are peer-reviewed and used at Google/Meta/Uber/Lyft. Brodersen et al. (2015) is the canonical Bayesian structural time-series paper.

Moderate evidence. Self-reported attribution captures real signal but is biased toward memorable/recent touchpoints.

Weak evidence. “90% measurement gap” as a universal claim reflects one agency’s SaaS sample, not a population estimate.

Plausible (a priori reasoning). Platform-owned successor metrics (ABS, QFC) carry their own structural biases toward the platforms that produce them. Same logic that applied to Apple’s SKAdNetwork applies here.

Section 7: Proposed Metrics

7.1 Leading indicators (in-platform)

- Reach, qualified impressions, frequency, frequency-capping evidence
- Watch time, completion rate, average view duration
- Saves, shares, substantive comments (≥ 10 words), DMs received
- Repeat exposure; audience overlap
- *Language adoption*: third parties repeating your framing/phrasing in posts (textual citation tracking via Brandwatch, Talkwalker)

7.2 Lagging indicators (with platform-source notes)

- Branded search volume (Google Trends + GSC + **ABS in Google Ads/GMP, May 2026+**)
- Direct traffic (much is dark social mis-classified per SparkToro 2022)
- **QFC (Qualified Future Conversions)**: 6-month horizon, Google-native; *use as one signal, validate against MMM and survey-based brand tracking*
- Newsletter signups, demo requests, pipeline created (CRM)
- Inbound unlinked brand mentions
- Self-reported attribution on conversion forms (“How did you hear about us”)
- Pipeline velocity; deal velocity; win rate; ACV
- Customer-cited reasons in sales call notes

7.3 Receipt metrics (qualitative)

- DM screenshots, comments quoting the brand
- Third-party blog posts, community mentions
- Reviews on G2 / Capterra / TrustRadius
- Reddit threads, podcast mentions
- LLM citation rates (Profound, Athena, Peec)

7.4 Reliability and biases

- *Reach inflation* (impression fraud, viewability $< 50\%$).
 - *Engagement bias*: outrage engagement $\neq$ preference (Nelson-Field).
 - *Branded-search confounds*: PR spikes, news cycle, competitor missteps.
 - *Self-report*: recency and emotional vividness.
 - *Receipt cherry-picking*: survivor / response bias.
 - **ABS/QFC walled-garden bias**: non-Google lift counted as Google.
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Section 8: GEO and Generative Search

8.1 Aggarwal et al. (2024): the academic anchor

Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24), pp. 5–16; arxiv:2311.09735. Authors: Pranjal Aggarwal (IIT Delhi), Vishvak Murahari (Princeton), Tanmay Rajpurohit (Independent), Ashwin Kalyan (AI2 / Independent), Karthik Narasimhan (Princeton), Ameet Deshpande (Princeton).

Key claims:

- Defines Generative Engines (GEs) that synthesize answers from multiple sources rather than rank links.
- Introduces **GEO-bench**, a large-scale benchmark of diverse user queries across domains.
- Demonstrates **up to 40% visibility boost** in generative-engine responses through black-box optimization of cited content.
- Visibility metrics include normalized word count in cited sentences, position in the answer, citation count (vs. simple rank in classic SEO).

Subsequent work (AutoGEO 2025, arxiv:2510.11438) reports gains up to 50.99% over the original “Fluency Optimization” baseline.

8.2 Four-mechanism model

1. **Retrievability**: crawlability, indexability, robots.txt, sitemaps, JavaScript rendering, accessibility. AI crawlers (GPTBot, ClaudeBot, PerplexityBot, Google-Extended) honor robots.txt directives.
2. **Extractability**: clear definitions, headings, citable claims with evidence proximate, statistics within context.
3. **Credibility**: proprietary data, named experts, transparent methodology, direct experience, customer stories.
4. **Public evidence**: third-party validation. Per Pew (2025), Wikipedia, YouTube, and Reddit account for ~15–17% of citations in both AI summaries and traditional search results.

Emerging empirical evidence on AI Overviews source selection. Xu, Iqbal, and Montgomery (2026), in a large-scale longitudinal measurement study of 55,393 trending queries across 19 topical categories (March 13 to April 21, 2026), report that AI Overviews activate on 13.7% of queries overall and 64.7% of question-form queries. Their most consequential finding for GEO research: nearly 30% of AIO-cited domains do not appear in the co-displayed first-page results, suggesting that source selection for generative answers operates through a layer distinct from Google’s classic ranking algorithm. This empirical observation supports treating GEO as related to, but methodologically distinct from, classic SEO ranking. The arXiv preprint status (not yet peer reviewed) is noted; the methodology is documented and the directional finding aligns with practitioner observations.

8.3 Google’s AI Search pivot and the limits of GEO framing

The most consequential confirmation for GEO practitioners came when Google’s product leadership explicitly laid out organic-visibility guidance for AI Search. Vidhya Srinivasan’s official guidance:

- **Uniqueness over commodity:** *“Lead with what your brand can uniquely say. Be specific. Stay away from generic, commodity information. You want one-of-a-kind content.”* (~47:13)
- **Firsthand evidence:** *“Share those reviews, share the firsthand experiences or expert takes that only you can provide.”* (~47:13)
- **Helpfulness over keyword-stuffing:** *“Focus on the most helpful content for your customers... Do not try to write for bots.”* (~47:39)
- **Multimodal readiness:** *“Think beyond text. People appreciate finding relevant images and videos. Generative AI search pulls all of this content right into the results.”* (~47:51)
- **Structured for agents:** *“Your content needs to be well structured, accessible and easy to navigate.”* (~48:02)

This moves the core principles of citable, structured, evidence-backed content closer to Google-recognized operational guidance. The Aggarwal et al. (2024) academic framing and Google’s product guidance now align on these underlying principles.

Important defensive note. Google’s position should not be read as an endorsement of every tactic marketed under “GEO” or “AEO.” In its generative AI search guidance, Google has explicitly stated that llms.txt files, AI-specific markup, content chunking strategies, and over-optimization of structured data are not required for visibility in Google’s generative AI features. The more defensible interpretation is that Google’s AI Search shift increases the importance of retrievable, useful, evidence-rich content (what AI search practitioners have variously called GEO, AEO, or “answer engine optimization”), not that it validates a separate set of vendor-marketed “GEO hacks.” Practitioners should treat Google’s AI Search principles as a refinement of long-known content-quality and technical-SEO fundamentals, not as a new doctrine sold under acronyms.

Google’s spam policies for Web Search also now explicitly include attempts to manipulate generative AI responses in Google Search as a category of spam, which further supports treating GEO as an evidence-quality and retrieval problem rather than a manipulation playbook (Google Search Central, Spam Policies, 2026).

8.4 Critical assessment of llms.txt

The **llms.txt** proposal (Jeremy Howard, Answer.AI, September 2024) is a Markdown standard placed at the site root to give LLMs a structured content map. Howard explicitly stated it is *“not an official standard”* and was designed for inference-time use by coding tools and agents, not for ranking in AI search.

Adoption reality (late 2025): BuiltWith recorded 844,000+ implementations; SE Ranking’s 300,000-domain study (November 2025) found 10.13% adoption rate; Rankability’s monthly scan of the top 1,000 global sites found zero proper imple-

mentations among major platforms. Critics call it a dud for GEO purposes; OtterlyAI removed its llms.txt check from their GEO audit.

Verdict: low evidence that llms.txt improves AI-search visibility today; useful for developer-tool integrations. Google did not endorse it at GML 2026.

8.5 Is GEO a new discipline?

Critical view: GEO is best characterized as an extension of SEO and content strategy with new measurement requirements. The Aggarwal et al. KDD 2024 paper is genuine peer-reviewed research; much of the surrounding industry discourse is consulting hype repackaging long-known content principles (clarity, citability, structured data via Schema.org). What changed at GML 2026 is platform-side legitimacy: Google's product leadership publicly emphasized overlapping principles on the keynote stage.

The honest evidence-based core: structured, citable, evidence-backed content with named authors and proprietary data is likelier to be quoted by LLMs and may also be more eligible for citation or retrieval in Google AI Search per platform-articulated guidance.

Section 9: Gated Content in a Zero-Click World

The empirical evidence on gating is mixed and largely industry-sourced:

- **Refine Labs (Chris Walker)** advocates ungating most content; HDYHAU data shows ungated thought-leadership outperforms gated whitepapers in self-reported attribution.
- **AI crawlers generally cannot pass forms.** Gated content is invisible to GPT-Bot, ClaudeBot, PerplexityBot, and Google-Extended. Gated content cannot enter the LLM corpus unless mirrored ungated.
- **B2B buyer-journey studies** (TrustRadius, 6sense, Gartner): 78% of buyers report selecting products they were aware of before research; ~90% choose from Day-1 shortlist (HBR).

Proposed rule (operational, not yet empirically proven): *Ungate the argument; gate the utility.* Leave open: thesis, viewpoint, framework, main findings, definition, comparison, citable idea. Gate only when there is genuine additional utility: template, dataset, deep benchmark, calculator, workshop registration, implementation guide.

Caveats. The rule is intuitive and aligns with the LLM-corpus argument but has not been tested in a randomized field experiment with conversion outcomes. Lead-volume vs lead-quality trade-offs may differ by ACV, sales cycle, and category maturity.

Section 10: Hypotheses (Status Updated)

#	Hypothesis	Status after GML 2026	Notes
H1	Complete native content generates greater comprehension than click-oriented teaser content	Untested in randomized lab setting	Open research question
H2	Comprehension mediates the relationship between exposure and recall	Untested in this domain	Open research question
H3	Repetition of a distinctive framing increases spontaneous recall	Untested in this domain	Open research question
H4	Brand recall increases the probability of subsequent branded search	Operationalized by ABS	Google now measures this directly within its ecosystem
H5	Direct traffic and branded search can capture demand created by previous non-attributable exposures	Operationalized by ABS + QFC	Same as H4, plus long-term horizon
H6	Click-based attribution underestimates distributed influence channels	Confirmed by platform owner	Rettke: <i>“this legacy concept of just clicks”</i> ; Google: 60% of conversions occur after 30 days
H7	Algorithmic platforms tend to favor content that keeps the user within the platform	Moderate-strong evidence	LinkedIn, X, Meta documented; Google’s AI Mode confirms
H8	Generative responses reduce some informational clicks but increase the importance of being cited or included in the response	Confirmed by platform owner	Google’s GEO guidance + AI Mode adoption data

#	Hypothesis	Status after GML 2026	Notes
H9	Gated content reduces participation in the public record if it hides information necessary for comprehension, citation, or discovery	Untested but mechanically supported	AI crawlers cannot pass forms
H10	Qualitative signals like DMs, call mentions, and framing repetition can act as early indicators of comprehension, recall, or preference	Untested	Open research question
H11 (new)	Platform-owned successor metrics to click-attribution (ABS, QFC) carry their own structural biases toward the platforms that produce them	A priori plausible; untested	Parallel to Apple's SKAdNetwork bias toward iOS lift attribution

Section 11: Future Research Agenda

(This section, originally a research design protocol, has been reframed as a research agenda. The protocol itself remains available as a separate document for future activation.)

11.1 Open research questions

1. **Lab experiment:** Does complete native content produce greater comprehension, recall, trust, and preference than teaser-with-link content? (Original H1-H6 of the protocol.)
2. **Field experiment:** Do native-content investments produce measurable lifts in ABS and QFC under controlled rollouts? Now feasible at lower cost because Google has built the measurement infrastructure.

3. **Methodological:** Do ABS and QFC systematically over-attribute to Google in environments with significant non-Google exposure (Meta, TikTok, podcasts, dark social)?
4. **Replication:** Replicate Aggarwal et al. (2024) GEO findings on production LLM engines (not benchmark only).
5. **Cross-cultural:** Replicate Pew/Ahrefs AIO-CTR analyses on non-English SERPs (Spanish, Portuguese, Hindi, Mandarin). Aral, Li, and Zuo (2026), based on 24,000 search queries across 243 countries (2.8M results in 2024-2025), document substantial cross-national variation in AI search exposure (Google AIO expanded from 7 to 229 countries over the year, with notable exclusions including France, Turkey, China, and Cuba) and report that AI search surfaces fewer long-tail sources and less response variety than traditional search. Future zero-click measurement work should account for this cross-national heterogeneity rather than assuming a single global model of AI-mediated influence.
6. **Longitudinal:** Track LLM-citation share against branded-search lift over 12-24 months.
7. **Sectoral:** Test 95-5 dynamic with category-specific repurchase intervals; B2C, B2B SaaS, medical/health services, education.
8. **Gating:** RCT comparing gated vs ungated thought-leadership on pipeline ACV.

11.2 Recommended research design (preserved from original)

A three-study mixed-methods design (PRISMA-ScR scoping review + pre-registered 4-arm RCT + geo-randomized field quasi-experiment with CausalImpact) remains the methodologically sound approach. The protocol for this design is available as a separate working document. ABS and QFC now reduce the measurement-stack-building burden for Study 3 substantially.

Section 12: Methodological Risks

- **Correlation vs causation.** Rising zero-click search correlates with AI Overviews launch but is not exclusively caused by it; mobile-first usage, voice, and Google’s own SERP features predate AIO.
- **Reach ≠ impact.** Per Nelson-Field, *“On average, only about 20% of impressions are actively seen, while 80% are wasted”; “85% of digital impressions last under 2.5 seconds,”* below the memory-formation threshold.
- **Engagement ≠ preference.** Outrage engagement does not equal brand preference.
- **Cherry-picking receipts.** DM screenshots are not random samples.
- **Survivorship bias.** Viral case studies omit thousands of failures.
- **Self-reported attribution bias.** Emotional and recent memories dominate.
- **Undue extrapolation.** SaaS B2B (Refine Labs sample) over-represented in dark-social discourse; B2C and emerging markets differ.
- **B2B vs B2C.** 95-5 dynamic applies to long-cycle purchases; mass-market FMCG operates closer to continuous in-market.

- **Inflating AI’s role.** Google’s own product narrative now amplifies AI Search’s importance; this is a category of source bias that did not exist in May 2025.
 - **Branded search $\neq$ pure brand effect.** PR spikes, news cycles, retargeting all inflate.
 - **Hawthorne effect / demand characteristics.** Knowing one is measured changes behavior; survey items can lead.
 - **Platform-owner-as-evidence-source bias (new).** When the platform owner becomes the loudest voice articulating the critique of the old paradigm AND the principal vendor of the replacement, careful research must distinguish “the narrative is correct” from “the platform owner has commercial reasons to amplify it.”
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Section 13: Conclusions and Claims-Evidence Table

13.1 Solid (high evidence)

- The share of Google searches resolving without an outbound click is large and rising; AI Overviews materially reduce CTR for informational queries.
- AI Mode and AI Overviews are large-scale phenomena (1B+ and 2.5B MAU respectively; doubling quarterly).
- Mere exposure, processing fluency, and mental availability are well-established mechanisms by which non-click impressions shape preference.
- 60:40 brand-to-activation allocation is one of the most replicated findings in marketing effectiveness.
- Click-based attribution systematically under-credits dark-social and out-of-platform exposure (now confirmed by platform-owner: “*this legacy concept of just clicks*”).
- The “ads as answers” model is the operative strategic direction of Google’s product line.

13.2 Plausible (moderate evidence)

- Native platform-resident content reaches more users than link posts on LinkedIn, X, and Meta surfaces.
- Being cited in AI Overviews / LLM answers is becoming a meaningful “position zero” with branded-search lift even when click yield is low.
- GEO interventions can increase visibility in generative answers; magnitude varies across domains.
- ABS and QFC will be useful causal-signal proxies for advertisers in 2026 and beyond.

13.3 Speculative

- The complete five-step Zero Click Ladder (Seen → Sought Out) as a single coherent causal chain.
- llms.txt as a driver of LLM-mediated discovery.

- “Ungate the argument, gate the utility” as an empirically optimal rule.
- Generalizing Refine Labs’ “97% dark social” finding to non-SaaS B2B or to B2C.
- The exact magnitude of platform-owned-metric bias in ABS and QFC.

13.4 Updated claims-evidence table

Claim	Evidence level	Source mix
Zero-click share of Google searches > 55% in US/EU	Strong	SparkToro 2024, Similarweb 2025, multiple panels
AI Overviews / AI Mode significantly reshape search behavior	Strong	Pew, Reuters Institute, Ahrefs, Seer, SISTRIX, Amsive, Semrush, BrightEdge, + Google GML 2026
AI Overviews reduce clicks ~50% (Pew) / 34-58% (Ahrefs) / 30% YoY (BrightEdge)	Moderate-Strong	Direction agreed; magnitude method-dependent
Click-based attribution under-credits long-cycle channels	Strong	Refine Labs + Google GML 2026 (40% of Demand Gen conversions in first 30 days)
Ads must function as answers, not redirects	Confirmed as platform direction	Vidhya Srinivasan: <i>“the best ads must be answers”</i>
Keyword paradigm is in decline as primary buying unit	Strong (as platform decision)	Srinivasan: <i>“You can’t choose keywords anymore”</i>
LinkedIn/X/Meta favor native content	Moderate-Strong	Owner statements (Musk, Mosseri), LinkedIn Engineering, Hootsuite/Van der Blom
60:40 brand-to-activation optimum (Binet & Field)	Strong	996 IPA cases; replicated 2018; convergent with Google’s 40%-in-30-days data
95-5 Rule (Dawes) generalizable	Moderate	Original B2B context; WARC reframings note category specificity
Mere exposure → liking	Strong	Bornstein 1989 meta-analysis, 208 studies, $r \approx 0.26$
Dark social drives $\geq 80\%$ of “demand creation” influence	Weak-Speculative	Single-agency SaaS sample, COI

Claim	Evidence level	Source mix
GEO content optimization yields up to 40% visibility lift	Moderate	Aggarwal et al. peer-reviewed KDD 2024; benchmark-based
GEO content principles (unique, firsthand, structured) improve AI-search visibility	Confirmed as platform-endorsed	Vidhya Srinivasan's organic guidance at GML 2026
llms.txt improves AI-search ranking	Weak/None	Inventor disclaims this; no major engine has endorsed
"Ungate the argument, gate the utility"	Speculative	Plausible, not empirically tested
Cross-channel MMM is the successor to multi-touch attribution	Confirmed as platform direction	Meridian + GA360 cross-channel integration announced at GML 2026
Long-term ROAS measurement is now possible at platform scale	Plausible	QFC announced with 6-month horizon; validation forthcoming
Platform-owned attribution successors (ABS, QFC) carry their own structural biases	Plausible (a priori)	Same logic as SKAdNetwork; not yet empirically demonstrated

Section 14: GML 2026 Primary Source Quote Inventory

14.1 Why this is a category shift

Until May 2026, the most defensible critic of the click-decline narrative was Google itself, which routinely disputed external CTR studies. At Google Marketing Live 2026, Google's own product leaders confirmed the empirical decline of the click-as-primary-metric AND announced new measurement infrastructure designed to quantify the very influence that click-attribution had been blind to. This is a category shift in evidence quality: the dispute is no longer between independent researchers and Google, it is now between Google's prior marketing communications and Google's own current product architecture.

14.2 Consolidated quote inventory

Vidhya Srinivasan, VP/GM Search Ads:

- "AI overviews... now has more than 2.5 billion monthly active users." (~28:24)
- "AI Mode... has already surpassed more than a billion monthly users, and we're seeing phenomenal growth. Searches in AI mode have been doubling every quarter since launch." (~29:04)

- *"AI mode searches on average three times as long as traditional searches."* (~32:03)
- *"Brainstorming searches are growing 30% faster than AI Mode searches overall."* (~32:30)
- *"It's time to acknowledge a fact. Google Search is AI Search through and through."* (~32:34)
- *"This means you can't think in terms of simple keywords anymore and you definitely can't manually choose them."* (~33:00)
- *"On average, at scale for every \$1 you spend on Google Search globally, you get an incremental \$6 back."* (~30:55)
- *"Lead with what your brand can uniquely say. Be specific. Stay away from generic, commodity information."* (~47:13)
- *"Share those reviews, share the firsthand experiences or expert takes that only you can provide."* (~47:13)
- *"Focus on the most helpful content for your customers... Do not try to write for bots."* (~47:39)
- *"Get your website ready for users, systems and agents. Your content needs to be well structured, accessible and easy to navigate."* (~48:02)
- *"Now you can ask Google anything, so the best ads must be answers."* (~50:51)
- *"You can't choose keywords anymore. Just let the best AI do it for you."* (~54:59)
- *"Your brand doesn't just show up; it actually answers a question."* (~54:45)

Nicky Rettke, VP YouTube Ads:

- *"Higher long-term ROAS is how you build a business that lasts, the difference between buying the click and investing in real growth."* (~1:20:25)
- *"Finally, providing modern multi-touch attribution that goes beyond this legacy concept of just clicks."* (~1:20:18)
- *"Only 40% of Demand Gen conversions happen in the first 30 days but really that's just the beginning."* (~1:21:31)
- Demand Gen + Search/PMAX: *"10% higher ROAS and 12% higher sales effectively"* (~1:09:23)

Gaurav Bhaya, VP Google Ads:

- *"First, Attributed Branded Searches now available in Google Ads and Google Marketing Platform. This is the clear signal that your ad created intent."* (~1:37:56)
- *"This is why we are introducing a new powerful metric: Qualified Future Conversions. They connect signals such as follow-up branded searches, video views or site visits up to six months in advance."* (~1:38:22)
- *"advertisers who build their 'data strength' with first-party data see an 11% average increase in incremental ROAS"* (~1:37:00)
- *"For the first time, you can analyze your performance across Google, TikTok, Pinterest, Snap and other social media platforms, all in one place."* (~1:39:35)

14.3 Case examples cited by Google (single-source, COI-flagged)

These are Google-reported case results without independent methodology disclosure. Treat as illustrative, not as evidence:

- IKEA: 65% lift in non-branded clicks, 28% boost in incremental ROAS (via AI Max).
- Lufthansa Group: 24% increase in ROAS (via AI Max).
- GM: 3x ROI on YouTube across the portfolio (via Demand Gen).
- Allianz: 4% incremental revenue lift (via Demand Gen).
- Petco: outperformed social benchmarks during Cyber Week (via Demand Gen).
- Doc Martens: 16% immediate revenue jump (first-party data integration).
- Crew Clothing: 70% uplift in long-term conversions (via QFC).

These are useful for narrative but are not independent evidence. Each is reported by Google in the context of selling its own products. Independent verification would require partner disclosure or third-party audit.

14.4 The Google paradox

Three observations on the structural tension in GML 2026's announcements:

14.4.1 Walled-garden capture of the new metrics

ABS and QFC are visible only inside Google Ads and Google Marketing Platform. Lift created by non-Google channels that drives branded search to a Google property will appear in ABS as if Google created the intent. This is the same playbook as Apple with SKAdNetwork: build the metric that measures the thing only you can see.

14.4.2 Removal of advertiser control

The shift from explicit keyword/audience targeting to conversational AI instructions ("AI Brief") concentrates targeting decisions inside Gemini's interpretation. From an experimenter's perspective, this is a measurement-darkening event: advertisers lose ability to run clean A/B tests on copy and audience because Gemini reinterprets both per delivery.

14.4.3 Universal Commerce Protocol

UCP, co-developed with Amazon, Meta, Microsoft, Salesforce, and Stripe, standardizes how agentic shopping flows interact with merchants. Framed as "open standard," its effect is to standardize commerce around the agent layer. For brands, the implication is loss of direct customer relationship in the discovery-to-transaction journey.

14.5 Secondary thesis introduced by GML 2026

The original investigation's primary thesis (the click is a valid demand-capture signal but insufficient as a sole proxy for marketing influence) is now strongly supported. GML 2026 introduces a secondary thesis worth tracking:

The alternative measurement infrastructure being built to replace click-attribution is being built primarily by the platforms that benefit from

controlling it. Marketers need both the new metrics (ABS, QFC) AND independent ways to validate them (MMM, geo-experiments, survey-based brand tracking, self-reported attribution).

This is the next research question, succeeding the original H7: not “does click-attribution underestimate influence?” (now confirmed), but “do the platform-owned successors to click-attribution introduce a different and equally severe bias toward the platforms that own them?”

Conflict of Interest Statement

This investigation acknowledges multiple categories of conflict of interest in its source base:

1. **Industry vendor sources** (SparkToro/Datos, Ahrefs, Similarweb, Refine Labs, BrightEdge, SISTRIX, Semrush, Seer, Amsive, Bain & Company): each has commercial interest in the narrative; their findings have been triangulated against independent and platform-owner sources where possible.
 2. **Platform-owner source (Google Marketing Live 2026)**: Google is simultaneously an evidence source AND a commercial actor with direct economic stake in the narrative. Every claim sourced from GML 2026 is simultaneously confirmatory of the zero-click thesis AND consistent with Google’s commercial interest in selling AI Max, Performance Max, ABS-based attribution products, and the Meridian-GA360 stack.
 3. **Independent academic and non-profit sources** (Pew Research Center, Reuters Institute, peer-reviewed papers): treated as the strongest evidence tier; explicitly distinguished from vendor and platform sources.
 4. **Author’s own conflicts**: The author, Emilio Alcolea, is the founder of Tersefy, an AI visibility and GEO agency for Tijuana cross-border specialty surgeons serving US patients. The author is the former Chief Sales & Marketing Officer at VIDA Wellness & Beauty Center (Tijuana) and Senior Marketing Consultant for Washington Vascular Specialists (USA). The Tersefy methodology referenced in this paper is built from operator work inside surgeon-led medical tourism practices, connecting AI visibility, entity architecture, content, reputation, and conversion systems. This investigation is published as a working paper rather than a peer-reviewed publication. Reader discretion is advised. No vendor or platform funded this investigation. No current or former client of the author is cited as a case example.
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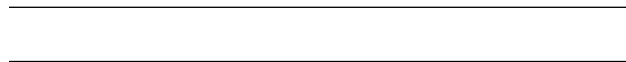
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- VCCP Media / Nelson-Field, K. (2025). Hacking the Attention Economy.

Tier 4: Platform-owner primary source (with explicit COI):

- **Google Marketing Live 2026** official broadcast, May 20-21, 2026. Key speakers: Vidhya Srinivasan (VP/GM Search Ads), Nicky Rettke (VP YouTube Ads), Gaurav Bhaya (VP Google Ads), John Nicoletti (VP Google Customer Solutions), Selin Song. Official keynote video: youtube.com/watch?v=T9585JkTPp8. Announcement page: blog.google/products/ads-commerce/ (Google Ads & Commerce Blog, May 20, 2026). All timestamps in this paper refer to the full keynote recording at the YouTube URL above, not to clipped uploads. *COI note: Google’s product leadership publicly articulated positions consistent with core claims of the click-decline thesis while simultaneously announcing Google-owned measurement products (ABS, QFC, Meridian-GA360 integration). Primary source with both corroborative value and direct commercial bias.*

Industry tools and frameworks:

- Meta Robyn (open-source MMM). github.com/facebookexperimental/robyn
- Google Meridian (open-source MMM, integrated into GA360 announced May 2026)
- Google CausalImpact (R / Python).
- LinkedIn B2B Institute. How B2B Brands Grow.



Appendix A: Full Evidence Matrix on Click Decline and Transformation

This appendix preserves the full evidence table referenced in Section 2.1 (simplified in body for readability).

Source	Year	Sample	Key finding	Method	COI flag
SparkToro/Dat2024 (Fishkin)		Multi-million-device Datos clickstream panel	58.5% US / 59.7% EU Google searches end without click; only 360 of 1,000 US searches go to open web	Panel clickstream, Sept 2022 - May 2024	SparkToro analyzes data from Datos (a Semrush company)
Pew Research Center	Jul 2025	900 US adults; 68,879 searches; 12,593 with AI Overviews	Click rate 8% with AI summary vs 15% without (46.7% relative reduction); 1% click citations inside AI summary; 26% vs 16% session termination	Browser- tracked Knowl- edgePanel; web- scraping replication April 7-17, 2025	Independent non-profit; Google disputed methodol- ogy
Ahrefs	Apr 2025	300,000 keywords	Position-1 CTR dropped 34.5% where AI Overviews present	GSC + Ahrefs Keywords Explorer	Ahrefs sells SEO tools
Ahrefs (update)	Dec 2025	300,000 keywords	Position-1 CTR drop deepened to 58%	Same	Same

Source	Year	Sample	Key finding	Method	COI flag
Similarweb	Jul 2025	News-query traffic	Zero-click news searches rose from 56% (May 2024) to ~69% (May 2025); organic news traffic fell from 2.3B to 1.7B visits	Clickstream	Similarweb sells analytics
Amsive	2025	700,000 keywords, 10 sites, 5 industries	Average CTR drop 15.49%; -19.98% on non-branded; -37.04% with featured snippets; +18.68% on branded	GSC analysis	Agency
Seer Interactive	2025	3,119 queries, 25.1M organic impressions	61% organic CTR decline; citation in AIO yields 35% more organic and 91% more paid clicks	Aggregated 42-client data	Agency
SISTRIX	2025	100M+ German keywords	Position-1 CTR on AIO-present SERPs 27% → 11% (-59%); estimated 265M lost monthly clicks in Germany	National panel	Sells SEO tools

Source	Year	Sample	Key finding	Method	COI flag
BrightEdge	May 2025	Post-AIO year measurement	CTR fell 30% YoY while impressions rose 49%; 8+ word queries 7× more likely to trigger an AIO than one year prior	Aggregated tracking	Sells SEO tools
Bain & Company	Feb 2025	“Consumer reliance on AI search results”	80% of consumers rely on AI for ≥40% of searches; 15-25% organic traffic reduction	Survey	Management consultancy
Reuters Institute	2025	~100,000 respondents across 48 countries	Social/video news 52% (2020) → 65% (2025); in US, social/video (54%) overtook TV (50%) and news websites (48%)	Survey	Independent (Oxford)

Source	Year	Sample	Key finding	Method	COI flag
Google (Vidhya Srinivasan, GML 2026)	May 2026	Platform- level	AI Overviews: 2.5B MAU; AI Mode: 1B+ MAU doubling every quarter; AI Mode searches 3x longer than traditional; “brainstorm- ing” queries grow 30% faster than AI Mode overall	Google’s internal data, disclosed publicly	Google has direct commercial interest in narrative
Google (Nicky Rettke, GML 2026)	May 2026	Demand Gen advertisers	Only 40% of Demand Gen conversions occur in the first 30 days	Google’s internal data	Same

Document Version and Replication

Version

- v3.1 EN, May 2026. Editorial pass: fixed Figure 1 caption duplication; corrected “Search SERPs” to “Search results” in figure; converted inline list-formatting issues in Section 2.2 to numbered lists with introductory text; restored proper bullet rendering across 18 list blocks; added “distributed influence” to keywords; softened residual tone in Section 1.4 (“publicly adopted parts of the underlying premise”), Section 6.4 (“broadly resembles”), and Section 8.3 (“moves... closer to”); replaced residual “platform-owner confirmation” in Section 2.3.
- v3.0 EN, May 2026. Major editorial revision incorporating peer feedback (rounds 1 and 2) on academic tone, citation rigor, and structural clarity. New additions: Figure 1 (distributed influence conceptual model), Table 2A/2B split (Zero Click Ladder), three new citations (Xu et al. 2026 arXiv on AIO measurement; Aral et al. 2026 arXiv on global AI search; Google Search Central guidance and spam policies). Tone changes: “platform-owner confirmation” reframed as “evidence consistent with directional thesis”; Section 6.4 “con-

cedes” reframed as “frames as legacy model”; Section 8.5 “endorsed exactly” softened to “publicly emphasized overlapping principles”; multiple micro-edits to defensive academic register. Em-dashes removed per author brand standard. Substantive evidence and conclusions unchanged from v2.0.

Replication

The full GML 2026 broadcast is publicly available. Direct quotes can be verified at the timestamps cited. Future iterations of this research should treat the broadcast as a single primary source with both evidentiary value and structural COI, similar to how SEC filings and earnings calls are treated in financial research.

This document is released as a working paper under the principle that systematic synthesis of a contested empirical claim, with explicit evidence-tier categorization and conflict-of-interest flagging, is more useful to practitioners and researchers than either uncritical vendor narratives or methodologically pristine but inaccessible academic papers.

Suggested citation

Alcolea, Emilio. (2026). Zero Click Marketing and the Measurement of Distributed Influence: Evidence from Search, Social Platforms, Generative AI, and Google Marketing Live 2026. Working paper, v3.1 EN. Tersefy.

Open data and materials

Evidence matrix, quote inventory, and source-link file are available upon request. A public OSF repository is planned for a future version.

Final note. This investigation is a synthesis at a moment in time (May 2026). The empirical claims are dated; the methodological principles are not. The recommended posture is methodological pluralism: triangulate clicks, brand metrics, self-reported attribution, MMM, and platform-owned causal signals. Treat “Zero Click Marketing” as a useful diagnostic frame, not a fully validated discipline. Remain critical of any single source, including the platform owner, that claims to settle the question.